



Yakeen Plus (2025)

Practice Test- 14

DURATION :200 Minutes

DATE : 12/12/2024

M.MARKS :720

ANSWER KEY

PHYSICS

1. (2)
2. (4)
3. (2)
4. (3)
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8. (3)
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50. (2)

CHEMISTRY

51. (4)
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ZOOLOGY

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199. (2)
200. (1)

SECTION-(I) PHYSICS

1. (2)

$$d\vec{v} = \vec{a}dt = (\hat{t}\hat{i} + 3t^2\hat{j})dt$$

$$\therefore \int_0^{\vec{v}} d\vec{v} = \int_0^{2s} (\hat{t}\hat{i} + 3t^2\hat{j})dt$$

or velocity after 2 s is,

$$\vec{v} = (2\hat{i} + 8\hat{j})\text{ m/s}$$

Speed after 2 s

$$v = \sqrt{(2)^2 + (8)^2} = \sqrt{68} \text{ m/s}$$

from work energy theorem

$$W = \frac{1}{2}mv^2 = \frac{1}{2} \times 68 = 34\text{J}$$

[New NCERT Class 11th Page No. 00]

2. (4)

A dimensionally incorrect equation may be correct.

[New NCERT Class 11th Page No. 00]

3. (2)

 $P \rightarrow$ Pressure $V \rightarrow$ Volume $T \rightarrow$ Temperature $R \rightarrow$ const

$$P = \frac{a}{V^2} ML^{-1}T^{-2} = \frac{a}{(L^3)^2} a = ML^{-1}T^{-2}L^6 =$$

$$[M^1L^5T^{-2}]$$

[New NCERT Class 11th Page No. 00]

4. (3)

$$[a] = [LT^{-2}]$$

$$[b] = [LT]$$

$$[c] = [T^2]$$

$$[u] = [LT^{-1}]$$

$$[a][T] = [LT^{-1}]$$

$$[a] = [LT^{-2}]$$

[New NCERT Class 11th Page No. 00]

5. (2)

$$F = at + bt^2$$

$$F \rightarrow N \rightarrow kg \frac{m}{s^2}$$

$$at \rightarrow N \rightarrow a.t : kg \frac{m}{s^2} ; a : kg \frac{m}{s^3} ;$$

$$\boxed{[a] : [MLT^{-3}]}$$

$$bt^2 \rightarrow N \rightarrow bt^2 : kg \frac{m}{s^2} ; b : kg \frac{m}{s^4} ; \boxed{[b] : [MLT^{-4}]}$$

[New NCERT Class 11th Page No. 00]

6. (1)

On changing the units of measurement of a physical quantity, the numerical value either decreases or increases. If the system of measurement is large, then the numerical value will be smaller if the system of measurement is smaller, then numerical value will be higher.

[New NCERT Class 11th Page No. 00]

7. (2)

Surface tension	Kg s ⁻²
Pressure	Kg m ⁻¹ s ⁻²
Viscosity	Kg m ⁻¹ s ⁻¹
Impulse	Kg m s ⁻¹

[New NCERT Class 11th Page No. 00]

8. (3)

Acceleration of the system and hence tension (T) on the string attached with the pulley is same in both the cases.

$$\text{Hence, } F_1 = F_2 = 2T$$

[New NCERT Class 11th Page No. 00]

9. (2)

$$t_{\text{ball}} = t_{\text{man}}$$

$$\frac{2u \sin \theta}{g} = \frac{9}{v} \quad (v = \text{velocity of man})$$

$$\therefore v = \frac{9g}{2u \sin \theta}$$

$$= \frac{9 \times 10}{2 \times 15 \times 0.6} = 5 \text{ m/s}$$

[New NCERT Class 11th Page No. 00]

10. (2)

$$\frac{1 \text{ kg m}}{s^2} = \frac{F(x \text{ kg})(ym)}{(zs)^2}$$

$$\therefore F = \frac{z^2}{xy}$$

[New NCERT Class 11th Page No. 00]

11. (3)

$$\text{Distance} = \frac{\pi r}{2} \text{ and Displacement} = r\sqrt{2}$$

$$\frac{\text{Displacement}}{\text{Distance}} = \frac{r\sqrt{2}}{\frac{\pi r}{2}}$$

$$= \frac{2\sqrt{2}}{\pi}$$

[New NCERT Class 11th Page No. 00]

12. (4)

$$P = \frac{a^3 b^2}{cd}$$

$$\frac{\Delta P}{P} \times 100\% = \left(\frac{3\Delta a}{a} + \frac{2\Delta b}{b} + \frac{\Delta c}{c} + \frac{\Delta d}{d} \right) \times 100\%$$

$$= 14\%$$

[New NCERT Class 11th Page No. 00]

13. (2)

$$LC = \frac{1 \text{ MSD}}{n} = \frac{0.1 \text{ mm}}{N+1}$$

$$LC = \frac{0.01 \text{ cm}}{N+1} = \frac{1}{100(N+1)} \text{ cm}$$

[New NCERT Class 11th Page No. 00]

14. (2)

Angular momentum

$$L = mvr$$

$$\text{Dimensional formula} = [\text{ML}^2\text{T}^{-1}]$$

$$\text{Torque } \tau = Fr$$

$$\text{Dimensional formula} = [\text{ML}^2\text{T}^{-2}]$$

$$\text{Stress } \sigma = \frac{F}{A}$$

$$\text{Dimensional formula} = [\text{ML}^{-1}\text{T}^{-2}]$$

$$\text{Pressure gradient } \left(\frac{dP}{dx} \right)$$

$$\text{Dimensional formula} = [\text{ML}^2\text{T}^{-2}]$$

[New NCERT Class 11th Page No. 00]

15. (1)

$$h = \frac{1}{2}gt^2 = \frac{1}{2} \times 10 \times (4)^2 = 80 \text{ m}$$

[New NCERT Class 11th Page No. 15]

16. (2)

$$\text{In time } T, h = \frac{1}{2}gT^2$$

$$\text{In time } \frac{3T}{4}, h' = \frac{9}{16} \left(\frac{1}{2}gT^2 \right)$$

$$= \frac{9h}{16} \text{ From ground} = h - h'$$

$$= h - \frac{9h}{16} = \frac{7h}{16}$$

[New NCERT Class 11th Page No. 16]

17. (4)

For free fall from a height $u = 0$ (initial velocity).

$$h = ut + \frac{1}{2}gt^2 = 0 + \frac{1}{2}gt^2$$

$$\Rightarrow \frac{h_1}{h_2} = \left(\frac{t_1}{t_2} \right)^2$$

$$\text{Given, } h_1 = 16 \text{ m, } h_2 = 25 \text{ m}$$

$$\therefore \frac{t_1}{t_2} = \sqrt{\frac{h_1}{h_2}} = \sqrt{\frac{16}{25}} = \frac{4}{5}$$

[New NCERT Class 11th Page No. 17]

18. (1)

The distance covered in 2s,

$$x = \frac{1}{2}a(2)^2 = 2a$$

The distance covered in next 2s,

$$y = \frac{1}{2}a(4)^2 - \frac{1}{2}a(2)^2 = 6a$$

$$\text{Now, } \frac{x}{y} = \frac{2a}{6a} = \frac{1}{3}$$

$$\Rightarrow y = 3x$$

[New NCERT Class 11th Page No. 17]

19. (3)

In uniform circular motion tangential acceleration remains zero. But magnitude of radial acceleration remain constant.

[New NCERT Class 11th Page No. 43]

20. (2)

The relative velocity of boat w.r.t. water

$$= v_{\text{boat}} - v_{\text{water}} = (3\hat{i} + 4\hat{j}) - (-3\hat{i} - 4\hat{j})$$

$$= 6\hat{i} + 8\hat{j}$$

[New NCERT Class 11th Page No. 42]

21. (2)

$$y = x \tan \theta \left(1 - \frac{x}{R} \right) \Rightarrow y = ax \left(1 - \frac{x}{a/b} \right)$$

$$\Rightarrow \theta = \tan^{-1}(a)$$

[New NCERT Class 11th Page No. 44]

22. (4)

$$R = \frac{u^2 \sin 2\theta}{g} = \frac{2u_x u_y}{g}$$

u_y is same for all.

$\therefore$ Range $\propto$ horizontal initial velocity (u_x)

In path 4 range is maximum so football possess maximum horizontal velocity in this path.

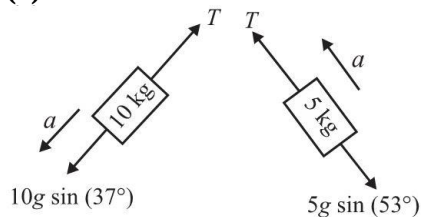
[New NCERT Class 11th Page No. 43]

23. (3)

$$\tan 30^\circ = \frac{a}{g} \Rightarrow a = \frac{g}{\sqrt{3}}$$

[New NCERT Class 11th Page No. 53]

24. (4)



$$10g \sin(37^\circ) - T = 10a \dots (i)$$

$$T - 5g \sin(53^\circ) = 5a$$

$$\text{Eq. (i)} - 2 \times \text{eq. (ii)}$$

$$\Rightarrow 10g \sin(37^\circ) - T - 2T + 10g \sin(53^\circ) = 0$$

$$\Rightarrow 3T = 10g \sin(37^\circ) + 10g \sin(53^\circ)$$

$$3T = 100 \times \frac{3}{5} + 100 \times \frac{4}{5}$$

$$T = \frac{700}{3} = 46.67 \text{ N}$$

[New NCERT Class 11th Page No. 00]

25. (2)

$$80 - 8g - 3g - 5g = (8 + 3 + 5)a$$

$$180 - 160 = 16a$$

$$20 = 16a$$

$$a = \frac{5}{4}$$

$$T - 3g - 5g = (3 + 5) \times \frac{5}{4}$$

$$T - 80 = 8 \times \frac{5}{4} = 10$$

$$T = 90 \text{ N}$$

[New NCERT Class 11th Page No. 00]

26. (3)

$$x = \frac{t^3}{2}$$

$$v = \frac{dx}{dt} = \frac{3}{2}t^2$$

$$\text{At } t=0, v_i = 0$$

$$\text{At } t=3 \text{ s,}$$

$$v_f = \frac{3}{2} \times 3^2 = \frac{27}{2}$$

According to work-energy theorem,

$$W = \Delta K = K_f - K_i$$

$$= \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$$

$$= \frac{1}{2} \times 1 \times \left(\frac{27}{2}\right)^2 = \frac{729}{8} \text{ J}$$

[New NCERT Class 11th Page No. 76]

27. (4)

$$\frac{1}{2}mv^2 = m \times g(50 - 10)$$

$$v = 20\sqrt{2} \text{ m/s}$$

[New NCERT Class 11th Page No. 75]

28. (4)

From energy conservation, $mgh \geq \frac{1}{2}mv^2$ at

lowest point ($v_{\min} = \sqrt{5gr}$ to complete vertical

circular motion) $mgh \geq \frac{1}{2}mv^2$

[New NCERT Class 11th Page No. 82]

29. (2)

Here,

$$U = \frac{A}{r^2} - \frac{B}{r}$$

For equilibrium, $\frac{dU}{dr} = 0$

$$\therefore -\frac{2A}{r^3} + \frac{B}{r^2} = 0$$

$$\text{or } \frac{2A}{r^3} = \frac{B}{r^2}$$

$$\text{or } r = \frac{2A}{B}$$

[New NCERT Class 11th Page No. 80]

30. (3)

$$\begin{aligned} \vec{r}_{\text{com}} &= \frac{1(\hat{i} + \hat{j}) + 2(2\hat{i} + 5\hat{j}) + 4(2\hat{i} - \hat{j})}{7} \\ &= \frac{\hat{i} + \hat{j} + 4\hat{i} + 10\hat{j} + 8\hat{i} - 4\hat{j}}{7} = \frac{13\hat{i} + 7\hat{j}}{7} \end{aligned}$$

[New NCERT Class 11th Page No. 96]

31. (1)

At maximum extension velocity of both the blocks will be same. Let v be the common velocity of the blocks (towards right). From conservation of linear momentum.

$$(3 + 6)v = 6 \times 2 - 3 \times 1 = 9$$

$$\text{or } v = 1 \text{ m/s}$$

Now let x be the maximum extension in the spring.

In From conservation of mechanical energy

$$\frac{1}{2}(3)(1)^2 + \frac{1}{2}(6)(2)^2 = \frac{1}{2}(200)x^2 + \frac{1}{2}(9)(1)^2$$

$$\text{or } 3 + 24 = 200x^2 + 9$$

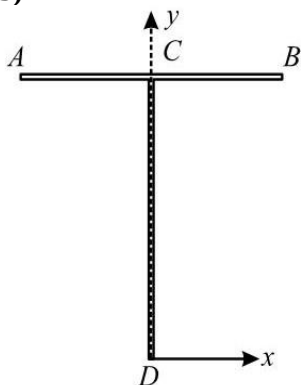
$$\text{or } x = \sqrt{\frac{18}{200}}$$

$$\text{or } x = 0.3 \text{ m}$$

$$\text{or } x = 30 \text{ cm}$$

[New NCERT Class 11th Page No. 96]

42. (3)

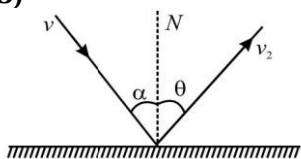


$$x_c = 0$$

$$y_c = \frac{m \times \frac{L}{2} + m \times L}{m + m} = \frac{3L}{4}$$

[New NCERT Class 11th Page No. 95, 96]

43. (3)



$$ev \cos \alpha = v_2 \cos \theta$$

$$v \sin \alpha = v_2 \sin \theta$$

$$\Rightarrow \frac{\tan \alpha}{e} = \tan \theta$$

$$\Rightarrow \theta = \tan^{-1} \left[\frac{\tan \alpha}{e} \right]$$

[New NCERT Class 11th Page No. 85, 86]

44. (4)

$$dm = \lambda dx = (A + Bx) dx$$

$$x_{CM} = \frac{\int x dm}{\int dm}$$

$$= \frac{\int_0^L (A + Bx) x \cdot dx}{\int_0^L (A + Bx) dx}$$

$$= \frac{\left[A \frac{x^2}{2} + B \frac{x^3}{3} \right]_0^L}{\left[Ax + \frac{Bx^2}{2} \right]_0^L}$$

$$= \frac{3AL + 2BL^2}{3(2A + BL)}$$

[New NCERT Class 11th Page No. 95, 96]

45. (1)

By conservation of linear momentum

$$mv = mv_1 + mv_2 \quad \dots (i)$$

$$\text{And } e = \frac{v_2 - v_1}{v - 0} \quad \dots (ii)$$

From (i) and (ii)

$$\frac{v_1}{v_2} = \frac{1 - e}{1 + e}$$

[New NCERT Class 11th Page No. 83]

46. (2)

$$x_{cm} = 0$$

$$y_1 = \frac{h}{3} = \frac{3}{3} = 1 \text{ m}$$

$$y_2 = h + \frac{3R}{8} = 3 + \frac{3}{8} \times \frac{8}{3} = 4 \text{ m.}$$

$$y_{cm} = \frac{M_1 y_1 + M_2 y_2}{M_1 + M_2} = \frac{1 + 4}{2}$$

$$= \frac{5}{2} \text{ m.}$$

[New NCERT Class 11th Page No. 95, 96]

47. (4)

$$F = -\frac{dU}{dx} = -(2x - 6)$$

For equilibrium, $F = 0$

$$x = 3 \text{ m}$$

$$\frac{d^2U}{dx^2} = 2x - 6 \quad \frac{d^2U}{dx^2} = 2 > 0$$

So, $x = 3$ is a point of stable equilibrium.

[New NCERT, Class 11th Page No. 77]

48. (1)

Here, $K_P > K_Q$

Case (1) : Elongation (x) in each spring is same.

$$W_P = \frac{1}{2} K_P x^2, W_Q = \frac{1}{2} K_Q x^2 \therefore W_P > W_Q$$

Case (2) Force of elongation is same.

$$\text{So, } x_1 = \frac{F}{K_P} \text{ and } x_2 = \frac{F}{K_Q}$$

$$W_P = \frac{1}{2} K_P x_1^2 = \frac{1}{2} \frac{F^2}{K_P}$$

$$W_Q = \frac{1}{2} K_Q x_2^2 = \frac{1}{2} \frac{F^2}{K_Q} \therefore W_P < W_Q$$

[New NCERT, Class 11th Page No. 80, 81]

49. (3)

A conservative force does not depend on the path along which the body is moved.

Example: Gravitational force, electrostatic force

In case of non-conservative force, the work done by force depends on the path along which the body moves and the work done by the force over a round trip is not zero

Example: Frictional force.

[New NCERT, Class 11th Page No. 81, 82]

50. (2)

$$\text{We can write, } \vec{s} = (4\hat{i} + 6\hat{j})$$

$$\text{And given } \vec{F} = (6\hat{i} - 8\hat{j}) N$$

And we can find work done as $W = \vec{F} \cdot \vec{s}$

$$\text{Now, Work} = (6\hat{i} - 8\hat{j}) \cdot (4\hat{i} + 6\hat{j})$$

$$\text{Work} = 24 - 48$$

$$\text{Work} = -24 \text{ J}$$

[New NCERT, Class 11th Page No. 74, 75]

SECTION-(II) CHEMISTRY

51. (4)

$$\Delta T_b = K_b \cdot m = K_b \cdot \frac{w_2}{M_2} \times \frac{1000}{w_1}$$

ΔT_b is a colligative property and it depends on the number of particles. Out of HF, HCl, HBr, HI, HI will have the lowest bond dissociation energy and will release a large number particles.

52. (1)

A solution of two volatile components showing negative deviation from Raoult's law shows decrease in vapour pressure and increase in boiling point. This is due to the greater adhesion than cohesion.

53. (1)

The solution having higher osmotic pressure than the reference solution is called hypertonic solution.

54. (4)

$$p = K_H \cdot \chi$$

K_H will be maximum if higher partial pressure is required to dissolve gas in liquid. A gas with weaker force of attraction will have maximum K_H .

55. (2)

H_3PO_3 is dibasic



56. (3)

$$\text{Molarity} = \frac{\text{number of moles of solute}}{\text{volume of solution (L)}}$$

Volume of a solution changes with the temperature and hence molarity.

57. (1)

Statement-I: When methyl alcohol is added to water, the boiling point of water decreases.

Statement-II: When a non-volatile solute is added to a volatile solvent, elevation in boiling point is observed.

58. (4)

Both sulphate and perchlorate ion have tetrahedral geometry.

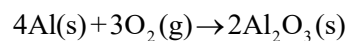
59. (3)

Given that an element with electronic configuration $[Ar]3d^1$ in its +2 oxidation state.

$$\text{The atomic number of element} = Ar + 1 + 2 = 21$$

Thus, the element is Scandium. It belongs to 4th period and group 3rd.

60. (2)

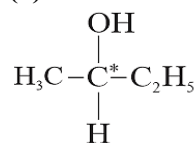


$$\text{Moles of Al} = \frac{4}{3} \times \text{moles of } O_2$$

$$= \frac{4}{3} \times \frac{1}{2} = \frac{2}{3}$$

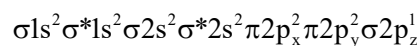
$$\text{Weight of Al} = \frac{2}{3} \times 27 = 18 \text{ g}$$

61. (3)



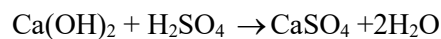
Optical active

62. (4)



Now from the electronic configuration we can see that the number of electrons in the $\sigma 2p_z$ molecular orbital is 1.

63. (1)



Ca(OH)_2 is Limiting reagent

1 mole of Ca(OH)_2 give

1 mole of CaSO_4 .

$\therefore$ 0.2 Mole of Ca(OH)_2 give

0.2 mole of CaSO_4 .

64. (3)

Both NF_3 and NH_3 have identical shapes and a lone pair of electrons on nitrogen. The dipole moment of NH_3 is higher than NF_3 due to different direction of moments of the N-F and N-H bonds. BF_3 is a symmetrical molecule. It has zero dipole moment.

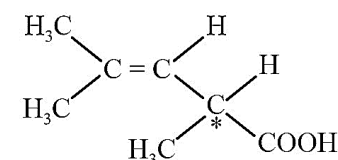
65. (1)

Acidity $\propto$ E.W.G

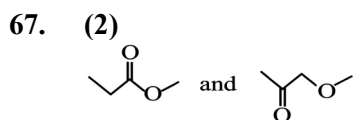
So, correct order of

acidity is: $A > B > C > D$

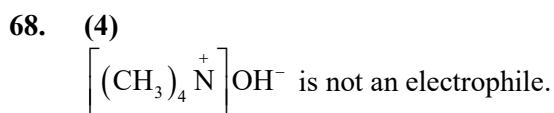
66. (2)



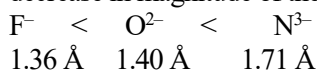
It exhibits optical isomerism.



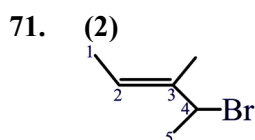
are functional isomers.



69. (1)
The ionic radii of isoelectronic ions increase with the decrease in magnitude of the nuclear charge.

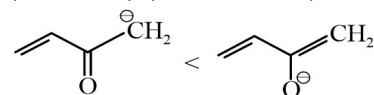
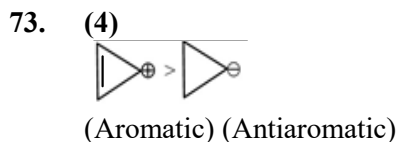
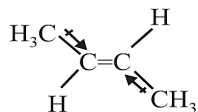


70. (3)
Stability of carbocation $\propto$ resonance effect or hyperconjugation effect or inductive effect.
The correct order is:
 $\text{I} > \text{III} > \text{II} > \text{IV}.$

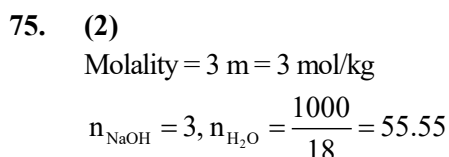
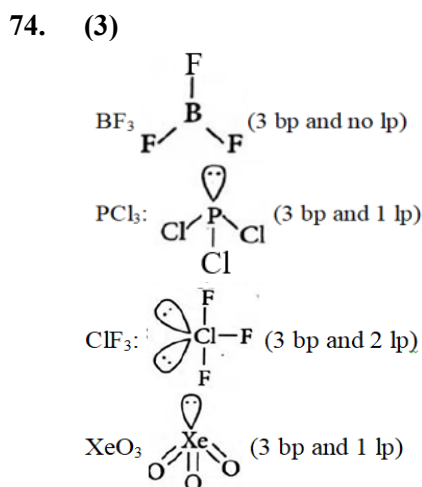


4-Bromo-3-methylpent-2-ene

72. (2)
Trans-But-2-ene has zero dipole moment.



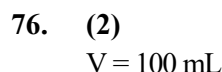
Negative charge is more stable on more electronegative atom.



$$\chi_{\text{NaOH}} = \frac{n_{\text{NaOH}}}{n_{\text{NaOH}} + n_{\text{H}_2\text{O}}} = \frac{3}{58.55}$$

$$\chi_{\text{NaOH}} = 0.05$$

$$\text{So, } \chi_{\text{H}_2\text{O}} = 0.95$$

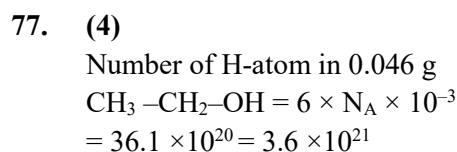


$$w_2 = 5.1 \text{ g}$$

$$M_2 = 34 \text{ g/mol}$$

$$M = \frac{\text{number of moles of solute } (n_2)}{\text{Volume of solution (in mL)}} \times 1000$$

$$= \frac{w_2}{M_2} \times \frac{1000}{V(\text{mL})} = \frac{5.1}{34} \times \frac{1000}{100} = 1.5 \text{ M}$$

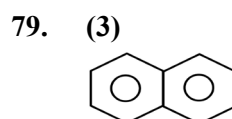


78. (3)

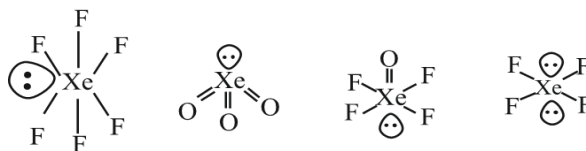
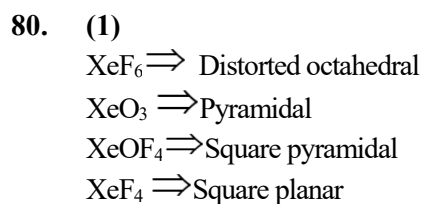
$$\text{Average atomic mass} = \frac{\sum \text{Percentage abundance} \times \text{isotopic mass}}{100}$$

$$= \frac{98.9 \times 12 + 1.1 \times 13}{100} = \frac{1186.8 + 14.3}{100}$$

$$= \frac{1201.1}{100} = 12.011 \text{ amu}$$



It is aromatic as it is planar, each carbon atom is sp^2 -hybridised and complete conjugation of $10 \pi e^-$.



81. (2)
Cyclopropane is a planar molecule and cannot form a conformational isomer.

82. (1)
Strength of the hydrogen bond is determined by the coulombic interaction between the lone-pair electrons of the electronegative atom of one molecule and the hydrogen atom of another molecule.

83. (3)
-
- Sec-pentyl alcohol neopentyl alcohol
- sec-pentyl alcohol and neopentyl alcohol are chain isomers.

84. (1)

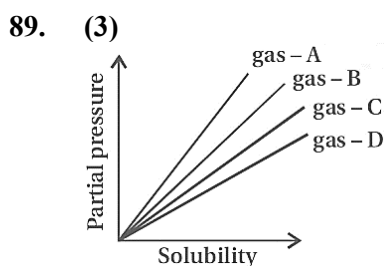
Species	Number of lone pairs
BF_4^-	12
BH_4^-	0
CN^-	2
COCl_2	8

85. (2)
- $\text{CH}_3 - \text{CH} = \text{CH} - \text{C} \equiv \text{N}$
 $\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $\text{sp}^3 \quad \text{sp}^2 \quad \text{sp}^2 \quad \text{sp}$

86. (2)
- The correct decreasing order of priority for functional group is
 $-\text{COOH}$, $-\text{SO}_3\text{H}$, $-\text{COOR}$, $-\text{COCl}$, $-\text{CONH}_2$, $-\text{CN}$, $-\text{NC}$, $-\text{CHO}$, $>\text{C}=\text{O}$, $-\text{OH}$, $-\text{SH}$, $-\text{NH}_2$

87. (4)
- The ebullioscopic constant of a liquid solvent is the elevation in boiling point of the solvent caused by the dissolution of 1 molal solution of a non-volatile and non-electrolyte solute in it.

88. (2)
- Equilibrium does not exist between solution and solute in a supersaturated solution.



$$\text{Slope } (K_H) = \frac{\text{Partial Pressure of gas}}{\text{Solubility of gas}}$$

In the graph, the slope (K_H) is highest for gas-A. At a constant partial pressure and temperature, the least solubility is of gas-A.

90. (1)
- $$\pi = CRT = \frac{5}{60} \times \frac{1000}{100} \times 0.0821 \times 300 = 20.5 \text{ atm.}$$

91. (2)
- $$\pi = CRT$$
- $$\text{Slope} = RT$$
- $$25.73 = 0.083 \times T$$
- $$T = \frac{25.73}{0.083} = 309.47 \approx 310 \text{ K}$$
- $$\therefore \text{Temperature in } ^\circ\text{C} = 310 - 273 = 37^\circ\text{C}$$

92. (4)
- Isotonic solutions at a particular temperature have same osmotic pressure.

$$\pi = iCRT$$

(A) 1 M NaCl	2M urea
$\pi = 2RT$	$\pi = 2RT$
(B) 1 M CaCl_2	1.5 M KCl
$\pi = 3RT$	$\pi = 3RT$
(C) 1.5 M KCl	2 M Na_2SO_4
$\pi = 6RT$	$\pi = 6RT$
(D) 2.5 M KCl	1 M $\text{Al}_2(\text{SO}_4)_3$
$\pi = 5RT$	$\pi = 5RT$

93. (4)
- SO_2Cl_2 NO
 $:\text{N}::\text{O}:$
Tetrahedral Paramagnetic
 NO_2^- I_3^-
 $:\ddot{\text{O}}::\text{N}::\ddot{\text{O}}:$ $[\text{I}-\text{I}-\text{I}]^-$
Diamagnetic Linear

94. (2)

19	s-block
22	d-block
32	p-block
64	f-block

95. (3)
- Resonance in O_3
Hydrogen bonding in HF
Covalent solid - C
Ionic solid - LiF

96. (1)

$1s^2 2s^2 2p^6$	+116 kJ/mol
$1s^2 2s^2 2p^6 3s^1$	-53 kJ/mol
$1s^2 2s^2 2p^5$	-328 kJ/mol
$1s^2 2s^2 2p^4$	-141 kJ/mol

97. (3)
-
- Pent-1-ene pent-2-ene
- Pent-1-ene and pent-2-ene are position isomers. Position isomers differ in the position of functional group or a substituent.

98. (4)
- No. of atoms present in a molecules of a gaseous element is called atomicity. So, atoms present in O_2 is 2. therefore, its atomicity is 2.
1 mole = 6.023×10^{23} . So, no. of atoms in 1 mole consists of 6.023×10^{23} atoms.

99. (3)
- Law of conservation of mass states that total mass of reactants is equal to total mass of product.

100. (3)
- $V \propto n$ (at constant T & P)
V = Volume of gas
n = Number of moles on number of particles.

SECTION-(III) BOTANY

101. (4)

In prokaryotes, ribosomes are associated with the plasma membrane of the cell. They are about 15 nm by 20 nm in size and are made of two subunits - 50S and 30S units which when present together form 70S prokaryotic ribosomes. Ribosomes are the site of protein synthesis. Several ribosomes may attach to a single mRNA and form a chain called polyribosomes or polysome. The ribosomes of a polysome translate the mRNA into proteins.

[New NCERT Class 11th Page No. 91]

102. (3)

Disintegration of nuclear membrane	Prophase
Appearance of nucleolus	Telophase
Division of centromere	Anaphase
Synthesis of RNA and proteins	G ₂ -phase

[New NCERT Class 11th Page No. 000]

103. (2)



These are white blood cell and are amoeboid in shape.

[New NCERT Class 11th Page No. 89]

104. (2)

The nucleoli are spherical structures present in the nucleoplasm. The content of nucleolus is continuous with the rest of the nucleoplasm as it is not a membrane bound structure. It is a site for active ribosomal RNA synthesis.

[New NCERT Class 11th Page No. 100]

105. (4)

Gas vacuoles are found in blue-green and purple and green photosynthetic bacteria.

[New NCERT Class 11th Page No. 91]

106. (2)

- Microtubules, microfilaments and intermediate filaments are collectively referred to as the cytoskeleton.
- Cytoskeleton in a cell are involved in many functions such as mechanical support, motility, maintenance of the shape of the cell.

[New NCERT Class 11th Page No. 98]

107. (1)

- Bivalent are tetrads.
- A bivalent means 4 chromatids and 2 centromere.
- One bivalent consists of pair of homologous chromosomes.
- Bivalents formation occurs in zygotene.

[New NCERT Class 11th Page No. 126]

108. (4)

If the initial amount of DNA is denoted as 2C then it increases to 4C. However, there is no increase in the chromosome number; if the cell had diploid or 2n number of chromosomes at G₁, even after S phase the number of chromosomes remains the same, i.e., 2n.

[New NCERT Class 11th Page No. 121]

109. (2)

Both belong to Kingdom plantae and division Angiospermae.

[New NCERT Class 11th Page No. 7, 8]

110. (4)

The **correct** sequence of phases of cell cycle is G₁ → S → G₂ → M.

[New NCERT Class 11th Page No. 121]

111. (4)

Recombination nodules are formed before diplotene during pachytene.

[New NCERT Class 11th Page No. 126]

112. (3)

These cells that do not divide further exit G₁ phase to enter an inactive stage called quiescent stage (G₀) of the cell cycle.

[New NCERT Class 11th Page No. 122]

113. (3)

In the S and G₂ phases the new DNA molecules formed are not distinct but intertwined.

Metaphase chromosome is made up of two sister chromatids.

[New NCERT Class 11th Page No. 122, 123]

114. (1)

During M phase, the number of chromosomes in the parent and progeny cells is the same, it is also called as equational division.

[New NCERT Class 11th Page No. 122]

115. (1)

Once perfect (sexual) stages of members of deuteromycetes were discovered they were often moved to ascomycetes and basidiomycetes.

[New NCERT Class 11th Page No. 18]

116. (3)

Common Name	Order	Class
Dog	Carnivora	Mammalia
Cat	Carnivora	Mammalia

P-Mammalia

Q- Carnivora

[New NCERT Class 11th Page No. 7]

117. (2)

- Numerical taxonomy is based on all observable characteristics.
- Cytotaxonomy is based on cytological information like chromosome number, structure and behaviour.

[New NCERT Class 11th Page No. 24]

118. (1)

Bacteria are complex in behaviour.

[New NCERT Class 11th Page No. 13]

119. (2)

Asexual reproduction is by the production of different types of spores, the most common being the zoospores.

[New NCERT Class 11th Page No. 24]

120. (4)

The first word denoting the genus starts with a capital letter while the specific epithet starts with a small letter. It can be illustrated with the example of *Mangifera indica*.

[New NCERT Class 11th Page No. 5]

121. (2)

The word systematics is derived from the Latin word 'systema'.

[New NCERT Class 11th Page No. 5]

122. (1)

Fusion of two nuclei is called karyogamy.

[New NCERT Class 11th Page No. 17]

123. (2)

The inner membrane of mitochondria forms a number of infoldings called the cristae (sing.: crista) towards the matrix.

[New NCERT Class 11th Page No. 97]

124. (4)

- Golden algae are unicellular eukaryotes.
- *Paramoecium* has gullet.
- *Gonyaulax* is dinoflagellate.
- Spores in slime moulds are dispersed by air currents.

[New NCERT Class 11th Page No. 15, 16]

125. (3)

Commonly known as sac-fungi, the ascomycetes are mostly multicellular, e.g., *Penicillium*, or rarely unicellular, e.g., yeast (*Saccharomyces*).

[New NCERT Class 11th Page No. 17]

126. (1)

Phycocerythrin is the major pigment in red algae.

[New NCERT Class 11th Page No. 27]

127. (2)

The number of species that are known and described range between 1.7-1.8 million.

[New NCERT Class 11th Page No. 3]

128. (4)

Ustilago → Smut

[New NCERT Class 11th Page No. 18]

129. (3)

Fusion between one large, nonmotile (static) female gamete and a smaller, motile male gamete is termed oogamous, e.g., *Volvox*, *Fucus*.

[New NCERT Class 11th Page No. 24]

130. (4)

International Code of Zoological Nomenclature (ICZN).

[New NCERT Class 11th Page No. 4]

131. (3)

Amyloplasts	Starch
Elaioplasts	Oils and fat
Aleuroplasts	Protein
Chromoplasts	Carotene

[New NCERT Class 11th Page No. 97]

132. (3)

At least a half of the total CO₂ fixation on earth is carried out through photosynthesis by algae.

[New NCERT Class 11th Page No. 26]

133. (3)

Classes	Common Name	Major Pigments	Stored Food
Chlorophyceae	Green algae	Chlorophyll <i>a, b</i>	Starch
Phaeophyceae	Brown algae	Chlorophyll <i>a, c</i> , fucoxanthin	Mannitol, laminarin
Rhodophyceae	Red algae	Chlorophyll <i>a, d</i> , Phycoerythrin	Floridean starch

[New NCERT Class 11th Page No. 27]

134. (1)

The main plant body of the bryophyte is haploid. It produces gametes, hence is called a gametophyte.

[New NCERT Class 11th Page No. 29]

135. (1)

Each genus may have one or more than one specific epithets representing different organisms, but having morphological similarities.

[New NCERT Class 11th Page No. 6]

136. (2)

In pteridophytes, the main plant body is a sporophyte which is differentiated into true root, stem and leaves.

[New NCERT Class 11th Page No. 30]

137. (1)

The interphase lasts more than 95% of the duration of cell cycle.

[New NCERT Class 11th Page No. 121]

138. (3)

Diakinesis is the final stage of prophase-I. Terminalisation of chiasmata occurs in diakinesis.

[New NCERT Class 11th Page No. 126]

139. (2)

Zoospores	Motile asexual spores in phycomycetes
Aplanospores	Non-motile asexual spores in phycomycetes.
Conidia	Asexual spores in deuteromycetes and ascomycetes
Basidiospores	Sexual spores in basidiomycetes

[New NCERT Class 11th Page No. 17, 18]

140. (4)

Algae are chlorophyll-bearing, simple, thalloid, autotrophic and largely aquatic (both fresh water and marine) organisms.

[New NCERT Class 11th Page No. 26]

141. (3)

Centrosome is an organelle usually containing two cylindrical structures called centrioles. They are surrounded by amorphous pericentriolar materials. Both the centrioles in a centrosome lie perpendicular to each other in which each has an organisation like the cartwheel.

[New NCERT Class 11th Page No. 99]

142. (2)

Two equatorial plates are formed during metaphase-I of cell cycle.

[New NCERT Class 11th Page No. 127]

143. (2)

Volvox, *Spirogyra*, *Chlamydomonas*, *Chara*, *Ulothrix* - Green algae
Polysiphonia, *Porphyra*, *Gracilaria*- Red algae
Fucus, *Ectocarpus* – Brown algae

[New NCERT Class 11th Page No. 26, 27, 28]

144. (3)

Certain marine brown and red algae produce large amounts of hydrocolloids (water holding substances), e.g., algin (brown algae) and carrageen (red algae) which are used commercially.

[New NCERT Class 11th Page No. 26]

145. (3)

In 1971, T.O. Diener discovered a new infectious agent that was smaller than viruses. It was found to be a free RNA; it lacked the protein coat that is found in viruses, hence the name viroid. The RNA of the viroid was of low molecular weight.

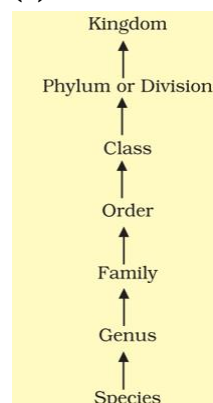
[New NCERT Class 11th Page No. 21]

146. (3)

Lion (*Panthera leo*) and leopard (*Panthera pardus*)

[New NCERT Class 11th Page No. 7]

147. (4)



[New NCERT Class 11th Page No. 8]

148. (4)

R.H. Whittaker (1969) proposed a Five Kingdom Classification. The kingdoms defined by him were named Monera, Protista, Fungi, Plantae and Animalia. The main criteria for classification used by him include cell structure, body organisation, mode of nutrition, reproduction and phylogenetic relationships.

[New NCERT Class 11th Page No. 11]

149. (3)

Lysosomes are membrane bound vesicular structures formed by the process of packaging in the Golgi apparatus. The isolated lysosomal vesicles have been found to be very rich in almost all types of hydrolytic enzymes. These enzymes are capable of digesting carbohydrates, proteins, lipids and nucleic acids.

The golgi apparatus principally performs the function of packaging materials, to be delivered either to the intra-cellular targets or secreted outside the cell. Golgi apparatus is the important site of formation of glycoproteins and glycolipids. Golgi complex and lysosome are part of endomembrane system.

[New NCERT Class 11th Page No. 95, 96]

150. (3)

The pili are elongated tubular structures made up of a protein.

[New NCERT Class 11th Page No. 91]

SECTION-(IV) ZOOLOGY**151. (1)**

Cuboidal epithelium	Ducts of glands
Ciliated epithelium	Fallopian tube
Compound epithelium	Moist surface of buccal cavity
Columnar epithelium	Wall of intestine

[Old NCERT Class 11th Page No. 103]**152. (2)**

Muscle fibres contract in response to stimulation, then relax and return to their uncontracted state in a coordinated fashion.

[Old NCERT Class 11th Page No. 103]**153. (4)**

Expiratory reserve volume is the additional volume of air, a person can expire by a forcible expiration. This averages 1000 mL to 1100 mL.

[New NCERT Class 11th Page No. 187]**154. (3)**

- The contraction of external intercostal muscles lifts up the ribs and sternum.
- The thoracic cavity is anatomically an air tight chamber.
- Healthy man can inspire approximately 6000 to 8000 of air per minute.
- During expiration, the intrapulmonary pressure is slightly higher than the surrounding atmospheric pressure.

[New NCERT Class 11th Page No. 186, 187]**155. (2)**

- The development of *Periplanata americana* is paurometabolous.
- Paurometabolous mean metamorphosis that involve nymphal stage.
- The embryo develop into a mature individual without involving a larval stage.

[Old NCERT Class 11th Page No. 115]**156. (4)**

Salivary gland is a multicellular gland.

[New NCERT Class 11th Page No. 102]**157. (2)**

Frogs are ureotelic animals.

[Old NCERT Class 11th Page No. 118]**158. (2)**

- Epithelial tissues form a protective barrier over body surfaces and line internal organs and cavities, preventing physical damage and infection.
- Epithelial tissues are involved in selective absorption, secretion, and excretion.

[Old NCERT Class 11th Page No. 101]**159. (1)**

Solubility of O₂ is 20-25 times lower than the CO₂.

[New NCERT Class 11th Page No. 105]**160. (4)**

The compound eyes of cockroaches are made up of about 2000 hexagonal ommatidia.

[Old NCERT Class 11th Page No. 114]**161. (3)**

Goblet cells	Present in alimentary canal
Adipose tissue	Fat storage
Endocrine gland	Hormone secretion
Salivary gland	Cluster of cells

[New NCERT Class 11th Page No. 102]**162. (1)**

Enzymes catalysing the linking together of 2 compounds, e.g., enzymes which catalyse joining of C-O, C-S, C-N, P-O etc. bonds are called ligases.

[New NCERT Class 11th Page No. 113]**163. (2)**

The cells of epithelial tissues are compactly packed with minimum intercellular matrix.

[New NCERT Class 11th Page No. 101]**164. (3)**

- In stone industries, so much dust is produced that the defense mechanism of the body cannot fully cope with the situation.
- Binding of oxygen with haemoglobin is primarily related to partial pressure of O₂.

[New NCERT Class 11th Page No. 191]**165. (1)**

In frogs, forebrain includes olfactory lobes, paired cerebral hemispheres and unpaired diencephalon.

[Old NCERT Class 11th Page No. 117]**166. (3)**

Neuron is the cell of nervous system.

[Old NCERT Class 11th Page No. 105]

167. (3)

Chitinous exoskeleton is found in insects.

[New NCERT Class 11th Page No. 44]

168. (3)

Both have placoid scales, a characteristic feature of Chondrichthyes

[New NCERT Class 11th Page No. 47, 48]

169. (4)

Triglycerides are the main constituents of body fat in humans and animals, as well as vegetable fat. They are formed from a single molecule of glycerol combined with three fatty acids on each of its hydroxyl (OH) groups. This structure makes them an important form of energy storage for many organisms.

[New NCERT Class 11th Page No. 106]

170. (1)



- *Nereis* belong to phylum Annelida that possess triploblastic, bilateral symmetry, metameric segmentation and coelom.
- *Nereis* is monoecious and has a closed circulatory system.

[New NCERT Class 11th Page No. 43]

171. (4)

In echinodermata, body cavity is lined by mesoderm.

[New NCERT Class 11th Page No. 45]

172. (4)

Petromyzon & Hagfish (*Myxine*) belongs to class cyclostomata, that lacks jaw.

[New NCERT Class 11th Page No. 47]

173. (1)

- Platyhelminthes have dorso-ventrally flattened body, hence are called flatworms.
- These are mostly endoparasites found in animals including human beings.
- Flatworms are bilaterally symmetrical, triploblastic and acoelomate animals with organ level of organisation.

[New NCERT Class 11th Page No. 42]

174. (1)

Aves (birds) and Mammals are warm-blooded animals, capable of maintaining a constant body temperature regardless of external environmental changes.

[New NCERT Class 11th Page No. 49, 50]

175. (4)

True coelom appeared first in the course of evolution in Annelida.

[New NCERT Class 11th Page No. 39]

176. (2)

In Urochordata, notochord is present only in larval tail, while in Cephalochordata, it extends from head to tail region and is persistent throughout their life.

Examples: *Urochordata* – *Ascidia*, *Salpa*, *Doliolum*; *Cephalochordata* – *Branchiostoma* (Amphioxus or Lancelet).

[New NCERT Class 11th Page No. 46]

177. (4)

Diaphragm is present in *Macaca*.

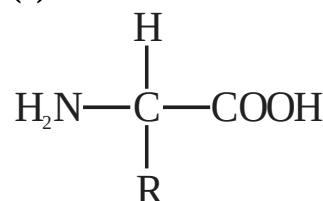
[New NCERT Class 11th Page No. 51]

178. (1)

Radial symmetry	Coelenterates
Pseudocoelomates	Aschelminthes
Metamerism	Annelida
Diploblastic	Sponges

[New NCERT Class 11th Page No. 38, 39]

179. (3)



A variable group designated as R group.

[New NCERT Class 11th Page No. 105]

180. (2)

Each nucleotide has three chemically distinct components. One is a heterocyclic compound (cyclic nitrogenous base), the second is a monosaccharide (pentose sugar) and the third is a phosphate group.

[New NCERT Class 11th Page No. 106]

181. (1)

- Inorganic catalysts work efficiently at high temperature and high pressure while enzymes get damaged at high temperature (40°C).
- Thermophilic organisms living in hot vents and sulphur springs have enzymes that are stable and retain their catalytic power even at high temperature (80 – 90°C).
- Ribozymes are nucleic acids behaving like enzymes.
- Malonate is a competitive inhibitor of succinate dehydrogenase.

[New NCERT Class 11th Page No. 113]

182. (1)

The tertiary structure of a protein is its three-dimensional conformation, which determines its specific biological function. If the tertiary structure is disrupted (e.g., by denaturation), the protein loses its functionality.

[New NCERT Class 11th Page No. 112]

183. (2)

In a nucleotide, the phosphate group is linked to the sugar (usually deoxyribose in DNA or ribose in RNA) via a phosphoester bond. This occurs when the hydroxyl group (-OH) of the sugar reacts with the phosphate group to form the ester linkage.

[New NCERT Class 11th Page No. 106]

184. (3)

- Lysine and arginine are basic amino acids as their side chain groups consist of a full positive charge at physiological pH.
- Lysine is moreover a basic and polar amino acid.
- As, the polar amino acid the side chain can interact with the water molecules.
- Lysine consists of two amino groups and one carboxylic group making it as basic in nature..(amino acids are basic).

[New NCERT Class 11th Page No. 104]

185. (3)

The acid-soluble pool contains small biomolecules with molecular weights 18 to around 800 Da.

[New NCERT Class 11th Page No. 105, 106]

186. (1)

Pressure of gases	Systemic veins	Systemic arteries
O ₂	40 mm Hg	95 mm Hg
CO ₂	45 mm Hg	40 mm Hg

[New NCERT Class 11th Page No. 187]

187. (1)

The respiratory control center described here is the pneumotaxic center, which is located in the pons and plays a role in moderating the activity of the respiratory rhythm center in the medulla oblongata. When the pneumotaxic center is absent or damaged, there is an increase in the depth of respiration.

[New NCERT Class 11th Page No. 190]

188. (4)

- The cell aggregate body plan is characteristic of phylum Porifera.
- Bilateral symmetry is the most common in higher animals, including vertebrates.
- Not all triploblastic animals have a true coelom. Some are acoelomates (e.g., Platyhelminthes), and others are pseudocoelomates (e.g., Nematoda)
- Platyhelminthes are acoelomates with no true body cavity.

[New NCERT Class 11th Page No. 38, 39]

189. (1)

- Relaxation of the diaphragm and the inter-costal muscles reduce the thoracic volume and thereby the pulmonary volume.
- This leads to an increase in intra- pulmonary pressure to slightly above the atmospheric pressure causing the expulsion of air from the lungs, i.e., expiration.

[New NCERT Class 11th Page No. 186]

190. (2)

The total number of ganglia in ventral nerve cord of cockroach is nine i.e., three in thoracic and six in abdominal.

[New NCERT Class 11th Page No. 114]

191. (2)

When CO binds to hemoglobin, it forms carboxyhemoglobin, which prevents oxygen from binding to hemoglobin and being transported to the body's tissues.

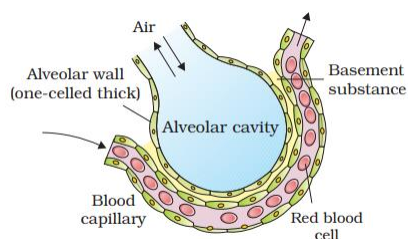
[New NCERT Class 11th Page No. 189]

192. (4)

Aquatic arthropods exhibit branchial respiration by the special vascularised structures called gills.

[New NCERT Class 11th Page No. 183]

193. (3)



[New NCERT Class 11th Page No. 188]

194. (1)

Tongue	Bilobed
Vasa efferentia	10–12 in number
Mesorchium	Double fold of peritoneum
RBCs	Nucleated

[New NCERT Class 11th Page No. 186, 187]

195. (3)

- Vital capacity- The maximum volume of air a person can breathe in after a forced expiration. This includes ERV, TV and IRV or the maximum volume of air a person can breathe out after a forced inspiration.
here, V.C. = Tidal volume + Expiratory reserve volume + Inspiratory reserve volume.
V.C. = 500 + 1000-1100 + 2500-3000
V.C. = 4000-4600 ml
- Tidal volume - Volume of air inspired or expired during a normal respiration. It is approx. 500 mL., i.e., a healthy man can inspire or expire approximately 6000 to 8000 mL of air per minute.
- When 1200 ml of air left in the lungs, it is termed as Residual Volume, i.e., Volume of air remaining in the lungs even after a forcible expiration.

- Inspiratory reserve volume – Additional volume of air, a person can inspire by a forcible inspiration. This averages 2500 mL to 3000 mL.

[New NCERT Class 11th Page No. 186, 187]

196. (3)

Coenzyme NADP contain vitamin niacin.

[New NCERT Class 11th Page No. 46]

197. (2)

In the alveoli, where there is high pO_2 , low pCO_2 , lesser H^+ concentration and lower temperature, the factors are all favourable for the formation of oxyhaemoglobin.

[New NCERT Class 11th Page No. 189]

198. (3)

Corvus, *Columba*, *Neophron* are aves while *Bufo* is amphibian.

[New NCERT Class 11th Page No. 48, 49]

199. (2)

- Bioluminescence is the property of a living organism to emit light, is well-marked in ctenophores.
- Some members of Platyhelminthes like *Planaria* possess high regeneration capacity.

[New NCERT Class 11th Page No. 42]

200. (1)

Palmitic acid has 16 carbons including carboxyl carbon. Arachidonic acid has 20 carbon atoms including the carboxyl carbon.

[New NCERT Class 11th Page No. 106]



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